Han Bao

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Personal Profile

Robotics researcher focusing on robot navigation and robot learning. My prior research tackled coupled motion planning and decision-making via deep reinforcement learning, while addressing implicit environmental perception through learning-based methods like contrastive learning. Currently, I am exploring how to adapt Vision-Language-Action(VLA) models for safe, kilometer-scale urban navigation. Additionally, my experience building a high-fidelity human-aware navigation benchmark has sparked a strong interest in RL-based character animation (DeepMimic, AMP) and humanoid control.

Education

2024 – present: MSc in Electrical and Electronic Engineering

Southern university of Science and Technology, Shenzhen
Supervised by Prof. Jiankun Wang.
Key Research Interests: Robot Learning, Robot Visual Navigation, Vision-Language-Action Model, Deep Reinforcement Learning for Robotics.

2020 – 2024: B.Eng in Robotics

Zhejiang University of Technology, Hangzhou
GPA: 3.83/5.00 (Rank: 1/31).
Relevant Coursework: Introduction to Robotics, Modern Control Theory, Principle of Automatic Control, Signals and Systems, Machine Vision.
Thesis: Research on Robot Motion Planning Methods Based on Deep Reinforcement Learning in Dynamic and Complex Environments.

Publications

Learning Robot Visual Navigation in Crowds via Intention-Aware Scene Representations.

IEEE Robotics and Automation Letters (RA-L 2026).(First author)
***Han Bao**, Bingyi Xia, Hanjing Ye, Yu Zhan, Hao Cheng, Baozhi Jia, Wenjun Xu, Jiankun Wang*

NavIsaacLab: Generating Realistic Crowd via Parallel Robot Learning for Benchmarking Human-aware Navigation.

Under review.(Equal contribution)
*Bingyi Xia, **Han Bao**, Jingyu Zhu, Hanjing Ye, Yuhan Pang, Guangcheng Chen, Liang Lin, Zhengyou Zhang, Wenjun Xu, Jiankun Wang*

Projects/Research

Robot Visual Navigation System in Dynamic Complex Environments | 2025

Shenzhen Key Laboratory of Robotics Perception and Intelligence (rπ Lab), SUSTech

- Designed an end-to-end DRL policy integrating a novel visual encoder for robot visual navigation in crowds, successfully achieving zero-shot sim-to-real deployment.
- Designed intention-aware scene representations that combine BEV-encoded dense occupancy features with attention-driven 3D pose analysis, effectively inferring human intentions to prioritize safe human-robot interactions.
- Built high-fidelity, human-centric Isaac Sim environments to train and benchmark human-aware navigation, outperforming current SOTA methods.
- **[Project Homepage]**

Vision-Guided Pan-Tilt Laser System | 2023

First Prize of Zhejiang Province, National Undergraduate Electronics Design Contest

- Developed a machine-vision-based pan-tilt control system that actuates a laser pointer for precise target contour tracing and real-time dynamic tracking of arbitrary moving objects.

Vision-Guided Self-Balancing Bicycle | 2022

National First Prize, 17th National Undergraduate Smart Car Contest (Balancing Bicycle Category)

- Engineered an autonomous self-balancing bicycle system driven by a reaction wheel for active posture control. By integrating a real-time machine vision pipeline for precise trajectory tracking, the platform achieved robust dynamic balancing and high-speed racing.
- **[Project Demo]**

Honors and Awards

- Outstanding Undergraduate Graduate of Zhejiang Province 2024
- Zhejiang Provincial Government Scholarship 2022, 2023
- First Prize (Zhejiang Province), National Undergraduate Electronics Design Contest 2023
- National First Prize, 17th National Undergraduate Smart Car Contest 2022

Skills

- Programming: Python, PyTorch, C/C++
- Robotic: ROS/ROS2, NVIDIA Isaac-Sim/IsaacLab

Academic Service

Journal & Conference Reviewer

IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Automation Science and Engineering (T-ASE)
IEEE International Conference on Robotics and Automation (ICRA)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)